

HelixVPN — Master Technical Specification (spine + index)

HelixVPN — Master Technical Specification (spine + index)

Revision: 1 **Last modified:** 2026-06-25T00:00:00Z **Status:** active — authoritative spine for the docs/research/mvp/final/ document set **Authority:** This document is the single entry point and architectural spine for HelixVPN. Every other document in final/ (00..11 + 99) is subordinate to and indexed from here. Where a subordinate document disagrees with this spine on scope, role definitions, the non-negotiable principles (§4), or the decision register (§9), this spine wins until amended per §11.4.73 (spec versioning). **Evidence base:** Synthesized from 16 source research documents under docs/research/mvp/ (11 LLM analyses 00, 01, 02, 03, 05, 06, 07, 08, 09, 10, 11 + the 5 refined 04_VPN_CLD documents). Citations use source ids: [04_ARCH \$N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_P0] = Phase0-Spike; [04_P1] = Phase1-MVP; [04_P2] = Phase2-Parity; [04_UI] = helix-ui-Flutter; [05_YB0] = mandated-stack brief; [02_QWN] [01_DSK] [07_GMI] [10_KMI] [11_MST] etc. = per-LLM analyses; [SYN] = cross-document synthesis.

0. How to read this specification

This is a **specification**, not the product. It describes *what to build and why*, to phase → task → subtask and near-code granularity. It does **not** contain the shipping implementation — that follows in the build phases. Two-to-three refinement passes will iterate on this set.

Reading order for a new engineer:

1. This spine — §1 (what it is) → §3 (roles) → §4 (principles) → §5 (one-screen architecture) → §8 (roadmap) → §9 (decisions).
2. Then the subordinate doc for your area, via the **INDEX (§10)**.
3. The original research for primary-source depth: start with [04_ARCH].

Conventions:

- **MUST / MUST NOT / SHOULD / MAY** carry RFC 2119 force.
- **Open decisions** are never silently resolved; they live in the **Decision Register (§9)** as *options + recommendation* per §11.4.66 / §11.4.101.

- Code blocks are *illustrative interface sketches* (traits, DDL, protobuf, FFI, skeletons), not final implementations.
-

1. Executive summary

HelixVPN is a **self-hostable overlay network with a privacy-VPN front end** [04_ARCH §1, SYN §1]. The one-sentence pitch: *Cloudflare Tunnel + WARP, rebuilt as Tailscale-style coordination, with Mullvad's obfuscation stack, fully self-hostable, on one shared codebase.*

The founding constraint [05_YBO, 00]: give remote users full, policy-scoped access to **one or many** internal / home / lab networks **without any inbound port-forward**. Internal hosts dial *outbound* to a public gateway (reverse tunnel); the gateway relays and routes. The headline differentiator over every incumbent: **one user → N joined private networks** (multi-network bidirectional gateway) with per-user ACL routing, *and* Mullvad-grade obfuscation including QUIC/MASQUE, *and* 8-platform reach on one codebase.

The cryptographic core is **WireGuard** everywhere; obfuscation/transport is a **pluggable layer beneath WireGuard** — never a fork of WG crypto [04_ARCH §2/§3]. Mullvad's "QUIC mode" is *not* a separate protocol — it is **WireGuard-over-MASQUE/HTTP-3** [04_ARCH §1, the single most important correction]. That decision drives the entire data-plane design.

The control plane is **Go + Gin + PostgreSQL (RLS) + Redis + Podman (rootless)** [05_YBO, mandated; confirmed by all analyses]. The client/connector/edge share a **Rust core (helix-core)**; all three apps share a **Flutter UI (helix-ui)**; all language clients are **schema-generated** from helix-*proto* so the codebases never drift [04_ARCH §5/§11].

Privacy is a **build property, not a config toggle**: there is no durable connection/traffic table; CI schema-lint fails the build if one appears [04_ARCH §7, 04_P1].

2. Product vision

Axis	Position
Primary buyer	The self-hoster / home-lab operator / small MSP who runs networks for several clients and wants Mullvad-grade privacy <i>and</i> Tailscale-grade coordination <i>without</i> a SaaS dependency [SYN §1].
Stance	Self-hosted / home-lab first (no-logs because <i>you</i> own it); the same code serves a managed/multi-tenant offering later. Licensing (source-available + commercial) is an open decision — see D8 (§9).
Lead differentiators	(1) self-hosted and Mullvad-grade obfuscation incl. MASQUE/QUIC; (2) 1 user → N networks bidirectional gateway with ACL routing; (3) genuine 8-platform reach incl. Aurora + HarmonyOS that no incumbent ships; (4) one shared

Axis	Position
	Rust+Flutter codebase; (5) event-driven real-time control plane (sub-second convergence) [04_ARCH §1.3].
Prior-art map	Mullvad (obfuscation bar), Tailscale/Headscale (coordination/network-map model), NetBird (closest OSS shape), Cloudflare Tunnel+WARP (connector-dials-out + MASQUE) — HelixVPN is the self-hosted union [04_ARCH §1.3].
Non-goals (MVP)	Browser system-wide VPN (impossible — TUN unavailable in a browser; web = Console + optional in-page WASM MASQUE proxy only [04_ARCH §5.7]); a public managed service (Phase 3+, decision-gated); novel crypto (WireGuard only).

3. The three roles and three apps

HelixVPN is defined by **three roles** on a reverse-tunnel hub-and-spoke topology, and **three first-party app classes** that drive them — all from one shared codebase [04_ARCH §1, SYN §1].

Role	Who runs it	Responsibility	Direction	Reuses
Connector	Network operator, <i>inside</i> a private network	Dials outbound to the gateway, authenticates, advertises served CIDRs , routes LAN traffic for authorized clients. The in-network half of "two-way".	Outbound-only	helix-core (advertise/route mode)
Gateway	The public VPS	Rendezvous hub. Runs control plane (Go) + data plane (Rust edge + kernel WireGuard). Authenticates, holds routing/policy truth, relays/routes between clients and the networks they may reach.	Accepts dials; never initiates into private nets	helix-core (edge consumer) + helix-go
Client	End user	Mullvad-style access app. Dials in, gets an overlay IP, reaches its authorized subset of joined networks, or uses the gateway as a plain privacy exit (full-tunnel).	Outbound-only	helix-core + helix-ui

App class	User	Primary jobs	Platforms
	End user		

App class	User	Primary jobs	Platforms
Helix Access		Connect, pick exit/network, toggle obfuscation, kill-switch, split-tunnel	iOS, Android, Aurora, HarmonyOS, Windows, Linux, macOS (Web limited) [04_ARCH §1.2]
Helix Connector	Network operator	Onboard a network, advertise CIDRs, set local ACLs, run headless	Linux/Windows/macOS daemon (+ optional Flutter UI); Android/embedded appliance
Helix Console	Admin	Tenants, users, devices, networks, routes, policies, audit, billing-optional	Web (responsive) + Desktop (same Flutter build; no helix-core , API client only)

All three share **one design system, one Dart UI core, one API client**. Access + Connector additionally share **one Rust VPN/transport core** [04_ARCH §1.2/§5.4].

4. Non-negotiable architectural principles

These are invariant across all phases. Violating any one is a redesign, not a tweak [04_ARCH §2, SYN §7].

1. **Control plane and data plane are strictly separated.** Go services **never** sit in the packet path. If the control plane is down, existing tunnels keep forwarding — **fail-static**.
2. **WireGuard is the cryptographic core; transports are pluggable.** Obfuscation is a swappable layer *under* WireGuard, never a fork of the crypto. (Noise IK, Curve25519, ChaCha20-Poly1305.)
3. **Outbound-only from edges.** Connectors and clients always dial the gateway. No private network ever needs an inbound hole.
4. **Push, don't poll.** State changes propagate over persistent channels as events; agents reconcile to a declared desired-state ("network map", Tailscale MapResponse-style). No cron-restart loops.
5. **One core per concern, reused everywhere.** Rust transport/VPN core shared by client, connector, *and* edge. Go domain libs shared across control services. Dart UI core shared across all three apps.
6. **Self-hostable by one person, scalable to many gateways.** One podman pod for a homelab; the *same images* scale to HA multi-region.
7. **No-logging by construction.** The data plane keeps only counters + ephemeral routing state. No connection/content logs. Privacy is a **build property**, enforced by CI schema-lint (§10 doc 10).
8. **Schema-first, zero drift.** Protobuf for agent contracts, OpenAPI for REST; Dart/Go/Rust clients are *generated* from helix-*proto*.
9. **Decoupled, reusable components** per constitution §11.4.28/§11.4.74 — each reuse pillar is its own vasic-digital repo, snake_case (§11.4.29), flat submodule (§11.4.28), with upstreams/ (§11.4.36); no nested own-org submodule chains.

5. One-screen architecture overview

5.1 C4-style component view

C4Container

```
title HelixVPN – System Container Diagram (C4 L2)

Person(enduser, "End user", "Wants privacy + access to joined networks")
Person(netop, "Network operator", "Owns a private LAN to expose")
Person(admin, "Admin", "Manages tenants, policy, devices")

System_Boundary(edge_net, "Private Network A/B (outbound-only)") {
    Container(connector, "Helix Connector", "Rust helix-core (advertise/route mode)", "Dials out, advertises CIDRs, routes LAN")
}

System_Boundary(gw, "HelixVPN Gateway (public VPS)") {
    Container(api, "API Gateway", "Go / Gin + Connect", "REST (apps) + gRPC (agents) + WS/SSE")
    Container(coord, "Coordinator", "Go, server-streaming", "Builds network maps, pushes deltas <1s")
    Container(ctrl, "Control services", "Go modular monolith", "identity / registry / ipam / pki / policy / telemetry")
    Container(edge, "Data-plane edge", "Rust (quinn+h3) + kernel WireGuard", "MASQUE termination, WG fast path, verdict maps")
    ContainerDb(pg, "PostgreSQL", "RLS multi-tenant", "Source of truth – NO connection/traffic tables")
    ContainerDb(redis, "Redis", "Streams + KV (TTL)", "Event bus + ephemeral presence/routing")
}

System_Boundary(apps, "Client devices") {
    Container(access, "Helix Access", "Flutter + Rust helix-core + tunnel shim", "Consumer VPN app")
    Container(console, "Helix Console", "Flutter (web+desktop), API client only", "Admin")
}

Rel(enduser, access, "Uses")
Rel(netop, connector, "Runs")
Rel(admin, console, "Uses")

Rel(connector, api, "Enroll + WatchNetworkMap", "gRPC/Connect over QUIC/H2 (outbound)")
Rel(access, edge, "WG over pluggable transport", "auto: UDP→LWO→MASQUE→SS→UoT")
Rel(access, api, "REST CRUD + WS/SSE", "live state")
Rel(console, api, "REST + WS/SSE", "admin")
```

```

Rel(api, ctrl, "in-proc calls")
Rel(api, redis, "XADD events")
Rel(coord, redis, "XREADGROUP + presence")
Rel(ctrl, pg, "RLS reads/writes")
Rel(coord, edge, "push verdict maps + routes")
Rel(edge, connector, "route packets to advertised CIDRs")

```

5.2 Data-plane layering (the obfuscation stack)

```

flowchart TB
    L4["L4 – Application traffic (user packets / advertised LAN)"]
    L3["L3 – WireGuard (Noise IK, ChaCha20–Poly1305) – CRYPTO CORE"]
    L2["L2 – Pluggable transport / obfuscation (exactly one active)"]
    L1["L1 – IP to gateway public endpoint"]
    DAITA["(orthogonal) DAITA shaping – constant sizing + cover traffic, ABOVE WG"]
    L4 --> L3 --> L2 --> L1
    DAITA -. applied above .-> L3
    subgraph TRANSPORTS ["L2 options"]
        direction LR
        T1["plain UDP (fast path)"]
        T2["MASQUE CONNECT-UDP / H3 (the QUIC mode)"]
        T3["Shadowsocks-wrap"]
        T4["UDP-over-TCP"]
        T5["LW0 (lightweight WG obfs)"]
        T6["port-hop / 443 / 53 disguise"]
    end
    L2 --- TRANSPORTS

```

The transport layer **only changes how the encrypted WG datagrams look on the wire**. Selection is automatic with manual override — the client tries plain WG, and after N failed handshakes escalates LW0 → MASQUE/QUIC → Shadowsocks → UoT (Mullvad's exact UX) [04_ARCH §3.2]. Full transport matrix and Transport trait → **doc 01-data-plane**.

5.3 Reuse map

```

flowchart LR
    subgraph core ["helix-core (Rust)"]
        transport["helix-transport (obfs)"]
        wg["helix-wg (WG control + boringtun fallback)"]
        ffi["helix-ffi (flutter_rust_bridge + UniFFI)"]
        recon["reconciler"]
    end
    subgraph ui ["helix-ui (Flutter/Dart)"]
        ds["helix_design system"]
        apiclient["api/ws client (generated)"]
        state["Riverpod state"]
    end
    proto["helix-proto (schemas)"]

```

```

core --> access2["Helix Access (client)"]
core --> connector2["Helix Connector"]
core --> edge2["Gateway edge (SAME crate, 3 consumers)"]
ui --> access2
ui --> connector2
ui --> console2["Helix Console (web+desktop)"]
proto --> dart["Dart client"]
proto --> go["Go client"]
proto --> rust["Rust client"]

```

Three reuse pillars: **Rust core** (client+connector+edge), **Flutter UI** (all apps), **schema-generated clients** (all langs) [04_ARCH §5.5].

6. Repository & component layout

Each top-level component is a **decoupled, independently reusable** unit → its own vasic-digital repo + flat submodule under the umbrella, snake_case, with upstreams/ recipes (§11.4.28/.29/.36/.74).

```

helixvpn/                                # umbrella
├─ helix_core/                            # Rust: WG control, transport,
reconciler, FFI      → reusable
├─ ── crates/helix_transport/            #   QUIC/MASQUE, Shadowsocks,
UoT, LW0  (client+edge)
├─ ── crates/helix_wg/                  #   WireGuard control +
boringtun fallback
├─ ── crates/helix_ffi/                 #   flutter_rust_bridge + UniFFI
surface
├─ helix_edge/                          # Rust: gateway data-plane edge
(uses helix_transport) → reusable
├─ helix_go/                            # Go control plane (modular
monolith)      → reusable
├─ helix_proto/                         # Protobuf + OpenAPI → generated
Dart/Go/Rust   → reusable
├─ helix_ui/                            # Flutter: design system +
screens + 3 flavors      → reusable
├─ ── app_access/  app_connector/  app_console/
├─ shims/                                # apple/ android/ windows/
linux/ harmonyos/ aurora/
├─ deploy/                              # Podman quadlets, Terraform,
Grafana-as-code, helixvpctl
├─ submodules/                          # containers, helix_qa,
challenges, docs_chain, security, vision_engine ...

```

Helix-ecosystem submodules that the source research predates and the spec **must** wire in [SYN §8]: containers (vasic-digital — the §11.4.76 mandated orchestration + on-demand test infra), helix_qa + challenges (anti-bluff QA/ Challenge layer §11.4.27/.5/.69/.107), docs_chain (spec-doc + workable-items sync §11.4.106), security (§7 tooling), vision_engine (video-evidence QA

§11.4.107/.158). Each wired or marked not-applicable-with-reason in **doc 05-repo-layout-tooling-and-helix-ecosystem** (deployment/orchestration integration) and **doc 10-testing-acceptance-and-qa** (helix_qa/challenges QA layer).

7. Cross-cutting contracts (defined once, referenced everywhere)

These three artifacts are the seams every component binds to. They are *summarized* here and *specified in full* in their owning docs.

7.1 The Transport trait (Rust, owning doc: 01-data-plane)

The single abstraction beneath WireGuard. One implementation, three consumers (client core, connector, edge) [04_ARCH §3.2].

```
/// helix_transport: every obfuscation mode implements this.
/// The WG engine sees only a datagram socket; the transport hides
/// how bytes look on the wire.
#[async_trait]
pub trait Transport: Send + Sync {
    /// Stable identifier used by the auto-ladder + telemetry
    /// (counts only, never content).
    fn kind(&self) -> TransportKind; // PlainUdp | MasqueH3 |
    Shadowsocks | UdpOverTcp | Lwo

    /// Establish the obfuscated carrier to `endpoint`. Returns
    /// when ready to pass WG datagrams.
    async fn connect(&self, endpoint: &Endpoint, cfg:
        &TransportCfg) -> Result<TransportConn, TransportError>;

    /// Server side (edge/connector): accept + de-obfuscate,
    /// yielding WG datagrams to the WG engine.
    async fn accept(&self, listener: &Listener) ->
        Result<TransportConn, TransportError>;
}

/// A live carrier: WG datagrams in/out, plus health the auto-
/// ladder reads.
pub trait TransportConn: Send + Sync {
    async fn send(&self, wg_datagram: &[u8]) -> Result<(),
        TransportError>;
    async fn recv(&self, buf: &mut [u8]) -> Result<usize,
        TransportError>;
    fn health(&self) -> CarrierHealth; // rtt, loss,
    handshakes_failed -> drives escalation
}
```


7.2 The WatchNetworkMap stream (protobuf, owning doc: 02-control-plane)

The push-based desired-state spine. Every agent opens it once; polling is abolished [04_ARCH §4.2/§4.4, 04_P1].

```
syntax = "proto3";
package helix.coordinator.v1;

service Coordinator {
    // Agent opens once; receives a full snapshot then a delta
    // stream. Peers are ALREADY policy-filtered
    // (need-to-know) – an agent never learns of peers it may not
    // reach.
    rpc WatchNetworkMap(WatchRequest) returns (stream
        NetworkMapEvent);
}

message WatchRequest {
    string device_id = 1;           // mTLS-bound; coordinator re-
    // checks identity
    uint64 known_version = 2;      // 0 = send full snapshot; else
    // resume from delta
}

message NetworkMapEvent {
    oneof body {
        Snapshot snapshot = 1;     // overlay_ip, allowed_peers,
        // routes, transport_policy, dns, killswitch
        Delta delta = 2;           // minimal change set for THIS
        // agent
    }
    uint64 version = 3;            // monotonic; convergence SLO: p99
    // < 1s
}
```

7.3 The FFI surface (helix-ffi, owning doc: 03-client-core-and-ui)

The Dart^[2]Rust boundary. Dart drives the core and consumes a status stream; UI state is a *pure function* of that stream [04_UI, 04_ARCH §5.1].

```
// helix_ffi (flutter_rust_bridge v2). Mirrors to UniFFI for
// native shims.
pub fn helix_start(cfg: ClientConfig) -> Result<(), HelixError>;
pub fn helix_stop() -> Result<(), HelixError>;
pub fn helix_set_exit(selection: ExitSelection) -> Result<(),
    HelixError>; // network | privacy-exit
pub fn helix_pin_transport(kind: Option<TransportKind>) ->
    Result<(), HelixError>; // None = auto-ladder
/// Hot stream the UI subscribes to; every state change pushes
/// here.
pub fn helix_status_stream(sink: StreamSink<HelixStatus>); //
    Disconnected|Connecting(TransportKind)|Connected|
    Reconnecting|Failed
```

8. Phase roadmap (the spine of phase → task → subtask)

Four phases, each with **hard exit gates**. A phase **MUST NOT** be declared complete until **every** gate passes with captured anti-bluff evidence (§11.4.5/.69/.107). Detailed task/subtask breakdowns live in the per-phase docs (00/01/.../11) and become **workable items** in the SQLite single-source-of-truth DB (§11.4.93/.95).

flowchart LR

```
P0["Phase 0 – Spike\n~3–4 wk, throwaway bodies\nnon production interfaces"] --> P1["Phase 1 – MVP\nself-hostable"]
```

```
P1 --> P2["Phase 2 – Parity + Reach\ndesktop, HA, full obfs, PQ"]
```

```
P2 --> P3["Phase 3 – Extended Reach\nHarmonyOS, Aurora, WASM, audit"]
```

8.0 Phase 0 — Spike (prove the hard parts) [04_P0]

Throwaway *bodies* on *production interfaces* — the surviving artifacts are the **interfaces**: the Transport trait, the helix-wg boringtun wrapper, the orchestrator + status enum, the FFI surface. Test rig: Linux netns + nftables DPI sim + tc netem loss/jitter. Milestones S0–S8.

Exit gates (ALL must pass):

Gate	Assertion	Evidence
G1	Plain-UDP WireGuard: client → gateway → connector LAN host, ≥80% of bare-link throughput	iperf3 capture, netns rig
G2	MASQUE/QUIC tunnel survives a DPI UDP block , ≥50% of plain-WG throughput	nftables block + throughput capture
G3	iOS NEPacketTunnelProvider runs the Rust core under the memory ceiling with ≥30% headroom (make-or-break — the reason core is Rust)	on-device RSS profile
G4	Go-vs-Rust edge benchmark decides D5 (MASQUE termination language)	head-to-head throughput/CPU/p99 capture
G5	flutter_rust_bridge FFI drives the core from Dart end-to-end	Flutter harness + status stream log
G6	Push-based reconcile from a static network map converges with zero polling	reconciler trace

Gate G3 failure ⇒ revisit D2 (core language); G2 failure ⇒ MASQUE library/transport rework before MVP. Owing doc: **00 + 06 + 02**.

8.1 Phase 1 — MVP (self-hostable) [04_P1]

Go modular monolith (identity / registry / ipam / pki / policy / coordinator / events / telemetry / api / store); Postgres + RLS (no connection/traffic tables — CI-lint enforced); protobuf
Coordinator.WatchNetworkMap over Connect; coordinator with in-mem topology graph + minimal deltas (**p99 convergence < 1s**); Redis Streams backbone; identity (OIDC + anonymous device tokens) + enrollment (device-generated WG keypair, **private key never leaves device**) + short-lived mTLS device cert + **revoke < 1s**; policy compiler (Tailscale-ACL-flavored, **default-deny, fail-closed** → AllowedIPs + nftables/eBPF verdict maps); Gin REST + WS/SSE; helixvpctl (Cobra) + Podman quadlets; hub-and-spoke data path; auto obfuscation ladder (plain / LWO / MASQUE).

Exit gate = the 8 MVP acceptance criteria (Definition of Done) [04_P1]:

1. Self-host from zero (one helixvpctl init on a clean VPS).
2. Enroll a connector **and** a client.
3. Client reaches an **authorized** LAN host **and** is **denied** an unauthorized one (default-deny proven).
4. **Auto-escalate to MASQUE** when plain WG is blocked.
5. Policy edit reflected < **1s, no restart**.
6. Device revoke enforced < **1s**.
7. Kill-switch + DNS-leak protection verified (no leak on tunnel drop).
8. **No durable connection log** exists (schema-lint green) **and** all three apps drive the system.

Owning docs: **00/01/03/04/05/06/07/09/10**.

8.2 Phase 2 — Parity + Reach [04_P2]

Full transport set (+Shadowsocks, UDP-over-TCP, hardened LWO; auto-ladder with **per-network memory + regional priors**); **DAITA via maybenot**; **direct P2P + NAT traversal** (STUN-like discovery, hole punching, DERP-style helix-relay fallback); **multi-hop** nested WG (entry/exit key separation); **post-quantum** handshake (ML-KEM/FIPS-203 PSK, **hybrid never PQ-only**; Rosenpass as alt); desktop apps (Windows wireguard-nt + privileged service; macOS Network Extension); policy-as-code / GitOps; HA + multi-region (stateless coordinators, Patroni Postgres, **NATS JetStream** — the D3 Phase-2 swap).

Exit gates: full Mullvad feature-parity matrix green [04_ARCH §6]; multi-hop + P2P + relay-fallback proven across a real NAT; PQ handshake interops + measured; failover < 30s with transport preserved; desktop apps pass the same 8 MVP DoD criteria.

8.3 Phase 3 — Extended Reach [04_P2 §12, 04_ARCH §12]

HarmonyOS NEXT (OpenHarmony SIG Flutter fork, Network Kit ability — biggest platform risk, real native tunnel-shim work) + **Aurora OS** (OMP Russia Flutter fork, Qt/C++ + tun; Russian-hosted toolchain → enterprise SKU); **WASM browser MASQUE proxy** (browser-scoped, *not* a system VPN); multi-tenant Console + audit + optional billing; **third-party security audit + reproducible builds**.

Exit gates: Access app builds + tunnels on HarmonyOS NEXT and Aurora (on-device evidence); web WASM proxy reaches a joined network from the browser; independent audit report + reproducible-build attestation.

9. Decision register (open decisions — options + recommendation)

Surfaced explicitly per §11.4.66 / §11.4.101 — **not** silently resolved. The refined 04_VPN_CLD docs pivoted to one camp; the broader 10-LLM consensus sometimes differs [SYN §3]. Each is a tracked workable item; the gate that resolves it is named.

#	Decision	Camp A	Camp B	Recommendation	Resolved by
D1	Primary obfuscating transport	MASQUE/QUIC = WG-over-HTTP-3 (RFC 9298/9297/9221) — Mullvad's <i>actual</i> mechanism [04_ARCH §3.3, 04_P0]	Hysteria2 + Salamander (turnkey QUIC+obfs) primary, WG fallback [plurality of 10 analyses]	MASQUE primary for true Mullvad parity + single Rust impl; keep Hysteria2 tuning knobs as quinn reference. Note: "Mullvad QUIC ≠ separate protocol, it IS WG-over-MASQUE" [04_ARCH §1].	doc 01-data-plane + gate G2
D2	Shared client-core language	Rust core + Flutter UI [04_*, plurality]	Go core + Flutter UI (reuses Hysteria2 Go) [01_DSK, 07_GMI]	Rust — wins on iOS NE memory ceiling + WASM. CLD = Rust.	G3
D3	Event bus	Redis Streams (MVP) → NATS JetStream (scale) [04_P1, 02_QWN, 11_MST]	NATS JetStream from start [01_DSK, 10_KMI]	Redis Streams MVP, NATS Phase 2 — taxonomy is bus-agnostic, swap is transport-only.	doc 02-control-plane
D4	IP-subnet collision across N RFC1918 nets	IPv6 ULA /48 per tenant + Tailscale 4via6 [04_ARCH §3.4]	CGNAT 100.64/10 1:1 per network [07_GMI, 10_KMI]	4via6 for scale + clean overlay; hide the mapping behind Console UX. A v1 must-decide .	doc 01-data-plane (addressing) / doc 04-security-privacy-pki (IPAM/policy)
D5	Gateway edge language (MASQUE termination)	Rust (quinn+h3, shares helix-transport)	Go (quic-go + masque-go turnkey)	Rust if G4 benchmark holds — single-implementation	G4

#	Decision	Camp A	Camp B	Recommendation	Resolved by
		byte-for-byte) [04_P0 G4]		guarantee; else Go acceptable.	
D6	Transport topology	single protocol end-to-end	asymmetric per-leg: Hysteria2/ QUIC user[?]gateway, WireGuard gateway[?]networks [11_MST]	Surface MST's best-fit-per-leg as a Phase-2 option; MVP keeps WG end-to-end with pluggable transport.	doc 01-data-plane
D7	MVP ambition	lean tunnel-first (CLI between 2 hosts → API → apps) [02_QWN, 01_DSK, 04_P0]	full ecosystem / "Connectivity-OS" from v1 [10_KMI, 03_ZAI]	Lean Phase-0 spike then MVP; defer the ecosystem to Phase 2/3.	§8 roadmap
D8	Licensing / positioning	source-available + commercial	pure OSS	Decide before public release [04_ARCH §13]; same code serves self-host + managed.	pre-Phase-3

10. INDEX — the final/ document set

This spine is SPECIFICATION.md. Every other document below is an actual sibling file under docs/research/mvp/final/. Each carries its own §11.4.44 revision header; .html/.pdf(/.docx) exports per §11.4.65/.153 + docs_chain sync (§11.4.106) are a deferred pass-2 deliverable (see REFINEMENT_NOTES.md). The **Glossary** is §11 of this spine; the **live Decision Register** is §9 of this spine; the **source-coverage ledger** is doc 99.

Doc	File	Scope	Owns
00	00-product-scope-and-principles.md	Product definition (is/is-not), personas, the three roles + three apps, two-way + multi-network model, self-host vs managed + licensing, in/out scope per phase, the 7 non-negotiable principles, the Mullvad-parity feature matrix, and the open decisions D1–D8 with recommendations	Requirements; principles; decision register
01	01-data-plane.md	Rust data plane: the Transport trait, transport matrix (plain UDP, MASQUE	D1, D6; the Transport trait

Doc	File	Scope	Owns
02	02-control-plane.md	CONNECT-UDP quinn+h3, CONNECT-IP, Shadowsocks, UoT, LWO, Hysteria2 option), helix-wg boringtun wrapper, orchestrator + status enum, auto-escalation ladder, routing (ULA /48, 4via6), default-deny ACL → AllowedIPs + nftables/eBPF, multi-hop nested WG, MTU Go modular monolith, full Postgres DDL + RLS, the protobuf Coordinator service + WatchNetworkMap streaming, coordinator brain (in-mem graph, minimal deltas, <1s SLO), Redis Streams backbone + taxonomy + XAUTOCLAIM DLQ, Gin REST + WS/SSE + Connect API surface, reconciliation flow	D3; data model; WatchNetworkMap; event bus
03	03-client-core-and-ui.md	helix-core FFI surface (start/stop/status_stream, TunnelStatus), Flutter helix-ui (helix_design system, Riverpod, 3 flavors via runHelixApp), per-platform TunnelPlatform shims (Apple NE, Android VpnService+JNI, Windows wireguard-nt+service, Linux, HarmonyOS, Aurora, Web-none), memory/size budgets, Melos monorepo	D2; FFI; UI/design system; platform shims
04	04-security-privacy-pki.md	Zero-trust default-deny, identity (OIDC + anon device tokens), enrollment + PKI (device keygen, key-never-leaves, short-lived mTLS, revoke<1s), key hierarchy/rotation, edge	Security/privacy invariants; PKI

Doc	File	Scope	Owns
		hardening, no-logging-as-code + CI schema-lint, kill-switch/DNS-leak state machine, post-quantum handshake, threat model	
05	05-repo-layout-tooling-and-helix-ecosystem.md	Monorepo layout, the planned decoupled vasic-digital component repos (deferred), schema-first codegen, helixvpctl, deployment as Podman quadlets + Docker Compose + Kubernetes manifests, and integration of the incorporated submodules/ (containers §11.4.76, helix_qa+challenges, docs_chain, security, vision_engine)	Repo layout; deploy; ecosystem
06	06-phase0-spike-wbs.md	Phase 0 Spike work-breakdown: gates G1–G6, milestones S0–S8 → tasks → subtasks (HVPN–P0–NNN), iOS-memory harness, Go-vs-Rust edge benchmark, netns+nftables-DPI+netem rig	Phase 0 execution plan
07	07-phase1-mvp-wbs.md	Phase 1 MVP work-breakdown (HVPN–P1–NNN), the 8-criterion MVP Definition-of-Done + SLOs as acceptance gates	Phase 1 execution plan
08	08-phase2-parity-wbs.md	Phase 2 Parity+Reach work-breakdown (HVPN–P2–NNN): full transport set, DAITA, P2P/NAT-traversal, multi-hop, PQ, desktop apps, GitOps, HA/multi-region	Phase 2 execution plan
09	09-phase3-reach-wbs.md	Phase 3 Extended-Reach work-breakdown (HVPN–P3–NNN): HarmonyOS NEXT, Aurora OS, WASM MASQUE proxy, billing-optional multi-tenant, third-party audit + reproducible builds	Phase 3 execution plan

Doc	File	Scope	Owns
10	10-testing-acceptance-and-qa.md	Anti-bluff QA strategy covering every §11.4.169 mandatory test type (unit/integration/e2e/full-automation/Challenges/HelixQA/DDoS/security/stress+chaos/concurrency/race/memory/benchmark), helix_qa+challenges integration, captured-evidence, per-phase acceptance gates, local-only enforcement (§11.4.156)	QA strategy; test-type coverage
11	11-deep-research-appendix.md	Cited external research per angle (wireguard, masque, hysteria2, mullvad, flutter_ffl, ios_android, go_cp, podman_k8s, pki_pq_nat, daita_test). Pass-1 status: facts marked UNVERIFIED — the web-research phase was rate-limited; pass 2 re-runs it (see REFINEMENT_NOTES.md R1).	External-source bibliography
99	99-source-coverage-ledger.md	Source-coverage proof: every one of the 16 source docs → which final/ doc(s) absorbed it + its unique contribution; honestly-disclosed gaps (§11.4.118)	Coverage proof
—	REFINEMENT_NOTES.md	Pass-1 → pass-2/3 punch-list: what's fixed, the open consistency worklist (R1–R6), and the deferred deliverables (repos, workable-items DB, exports)	The refinement plan

11. Glossary

Term	Definition
Connector	First-party agent inside a private network; dials outbound to the gateway, advertises CIDRs, routes LAN traffic for authorized clients [04_ARCH §1.1].

Term	Definition
Gateway	Public VPS running control plane (Go) + data-plane edge (Rust + kernel WG); the rendezvous hub.
Client	End-user access app; gets an overlay IP, reaches authorized networks or uses the gateway as a privacy exit.
Overlay IP	Stable per-node address in the tenant overlay (ULA IPv6 / 48 per tenant; advertised IPv4 LANs mapped via 4via6) [04_ARCH §3.4].
Network map	Per-agent desired-state document (overlay IP, <i>policy-filtered</i> peers, routes, transport policy, DNS, kill-switch posture) pushed over WatchNetworkMap [04_ARCH §4.4].
MASQUE	RFC 9298 CONNECT-UDP over HTTP/3 — the mechanism that wraps WireGuard datagrams to look like web traffic. The "QUIC mode" [04_ARCH §3.3].
LWO	Lightweight WG obfuscation — cheap XOR/padding scheme to evade naive WG signature blocks [04_ARCH §3.2].
DAITA	Defence Against AI-guided Traffic Analysis — constant packet sizing + cover traffic applied <i>above</i> WG (maybe-not-style state machine) [04_ARCH §3.1/§6].
4via6	Tailscale-style scheme mapping advertised IPv4 LANs into the tenant's ULA IPv6 space so overlapping RFC1918 ranges never collide [04_ARCH §3.4].
Auto-ladder	Client transport escalation: plain UDP → LWO → MASQUE/QUIC → Shadowsocks → UoT, driven by handshake-failure events [04_ARCH §3.2].
fail-static	Property that existing tunnels keep forwarding when the control plane is down [04_ARCH §2].
Verdict map	nftables/eBPF map compiled from the policy that grants/denies per-peer reachability at the edge [04_ARCH §3.4].
RLS	PostgreSQL Row-Level Security — enforces tenant isolation at the database, not just the app layer [04_ARCH §4.5].
No-logging-as-code	Architectural guarantee: no durable connection/traffic table exists; CI schema-lint fails the build if one appears [04_ARCH §7].
helix-core	Rust data-plane core shared by client, connector, and edge (WG control, transport, reconciler, FFI).
helix-transport	Rust crate implementing the Transport trait — one obfuscation implementation, three consumers.
WatchNetworkMap	Server-streaming RPC: snapshot + delta stream of desired state; replaces all polling [04_ARCH §4.2].
PWU	Parallel Work Unit — a self-contained workable item in the §11.4.58/.93 execution model.

12. Provenance & change control

- **Sources:** 16 research documents under docs/research/mvp/ (see citation key in metadata block). Primary architectural source: [04_ARCH]. Cross-cutting synthesis: [SYN] (v09-research/_SYNTHESIS.md).
- **Versioning (§11.4.73):** primary version bumps for major rewrites (this is V1); Revision bumps for additive refinements. This spine is Revision 1.
- **Sync (§11.4.106):** .html/.pdf exports of this and every final/ doc kept in lockstep via docs_chain.
- **Decision discipline (§11.4.66/.101):** open decisions live in §9 + doc 99, never silently resolved.
- **Anti-bluff (§11.4.5/.69/.107):** every phase exit gate requires captured evidence; no gate passes on metadata/config-only signal.

End of master spine. Subordinate documents 00–11 + 99 elaborate each area to phase → task → subtask and near-code granularity.